

Causal Temporal Post-Processing of Multimodal Detections for Aerial Search and Rescue

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Abstract—Autonomous unmanned aerial vehicles (UAVs) in wilderness search and rescue (SAR) require reliable person detection under strict size, weight, power, and computing (SWaP-C) limits. Combining visible red-green-blue (RGB) and thermal infrared (TIR) cameras prevents sensor-specific dropouts caused by canopy shadows, darkness, or thermal crossover. However, per-frame object detectors often produce momentary false alarms from terrain clutter. During autonomous flight, each false alarm triggers unnecessary loiter maneuvers, depletes battery reserves, and wastes telemetry bandwidth over constrained links. This paper evaluates lightweight causal temporal decision methods to generate reliable search alerts from multimodal aerial detections. Using a controlled multimodal benchmark spanning three biomes (arid desert, temperate forest, and snow alpine) across synchronized RGB and pseudo-TIR streams, we deploy two upstream You Only Look Once (YOLO11n) detectors (2.6M parameters, 6.5 giga floating-point operations [GFLOPs]) held strictly fixed. Confidences are combined through a late fusion max gate and filtered over a causal sliding window ($W = 5$ frames, 208 ms buffer latency bound at 24 Hz). We compare six temporal decision strategies with validation-calibrated thresholds: raw thresholding, moving average, median filtering, history consensus, a learned Mamba selective state-space model (SSSM), and a gated recurrent unit (GRU) baseline. On alert-level metrics, temporal post-processing establishes distinct operational trade-offs: a five-frame moving average achieves the lowest false alert rate among non-learned methods, reducing false alerts from 9.0 to 2.1 false alerts per minute (FA/min) (a 77% reduction, 65.3% FASR) with 84.6% target recall (11/13 tiles), while five-frame median filtering preserves 92.3% recall (12/13 tiles) with faster 291.7 ms time-to-alarm at 3.6 FA/min. Learned sequence models do not improve alert reliability over moving averages: Mamba-SSSM yields statistically indistinguishable false alerts ($p = 0.4922$) while GRU triggers significantly more false alerts ($p = 0.0178$), both incurring $73\times$ to $500\times$ higher execution latency. Lightweight filtering thus provides an optimal, training-free layer for edge avionics.

Index Terms—Multimodal fusion, search and rescue, unmanned aerial vehicles, state-space models, temporal filtering.

I. INTRODUCTION

Autonomous unmanned aerial vehicles (UAVs) provide rapid aerial coverage in wilderness search and rescue (SAR) missions [1], [2]. However, onboard vision systems must operate under strict size, weight, power, and computing (SWaP-C) limits. Remote missions depend on low-bandwidth satellite or cellular Internet of Things (IoT) telemetry [3]. In autonomous

operations, false alarms trigger costly hover-and-verify maneuvers that drain battery power and risk dispatching rescue teams to empty locations.

Wilderness optical sensing suffers from sensor-specific failure modes. Visible red-green-blue (RGB) cameras degrade in dense shadows, heavy foliage, and poor lighting. Thermal infrared (TIR) cameras detect body heat regardless of illumination [4], [5], but suffer from thermal crossover when ground temperatures match human body temperature [6]. In desert environments, sun-heated rocks emit thermal patterns that mimic human signatures. In snowy terrain, bright reflective snow causes visible glare, while insulated winter clothing reduces body heat emission. Combining visible RGB and pseudo-thermal sensors is therefore motivated by the complementary failure modes observed across diverse biomes.

Small UAVs require compact single-stage detectors such as You Only Look Once (YOLO) [7]. In this work, we deploy Ultralytics YOLO11n [8] (2.6M parameters, 6.5 giga floating-point operations [GFLOPs] at 640×640) trained in 32-bit floating-point (FP32) precision. However, per-frame detections fluctuate due to platform vibration, motion blur, and background clutter. These fluctuations create brief false alarms in background areas and intermittent dropouts on actual human targets.

To address these issues, we evaluate lightweight causal temporal decision methods that convert noisy per-frame multimodal detections into dependable mission alerts. Upstream detectors and late fusion remain strictly fixed across all experiments to isolate the impact of temporal filtering. Frame detections are combined using a late fusion max gate and filtered across a causal five-frame sliding window (208 ms at 24 Hz). We benchmark six temporal decision strategies under validation-calibrated thresholds: raw thresholding, moving average, median filtering, history consensus, a learned Mamba selective state-space model (SSSM) [9], and a learned gated recurrent unit (GRU) baseline.

We formulate the central hypothesis: *lightweight causal temporal post-processing substantially reduces false alerts from multimodal aerial detections while preserving high mission target recall, and investigate whether added sequence modeling complexity (recurrent gating or selective state-space modeling) provides measurable operational benefit over linear finite impulse response and order-statistic filtering.*

This work provides three main contributions:

- 1) **Multimodal SAR Benchmark:** A controlled 3×2 ex-

This work involved human subject detection in simulated imagery. The author(s) confirm(s) that all human/animal subject research procedures and protocols are exempt from review board approval.

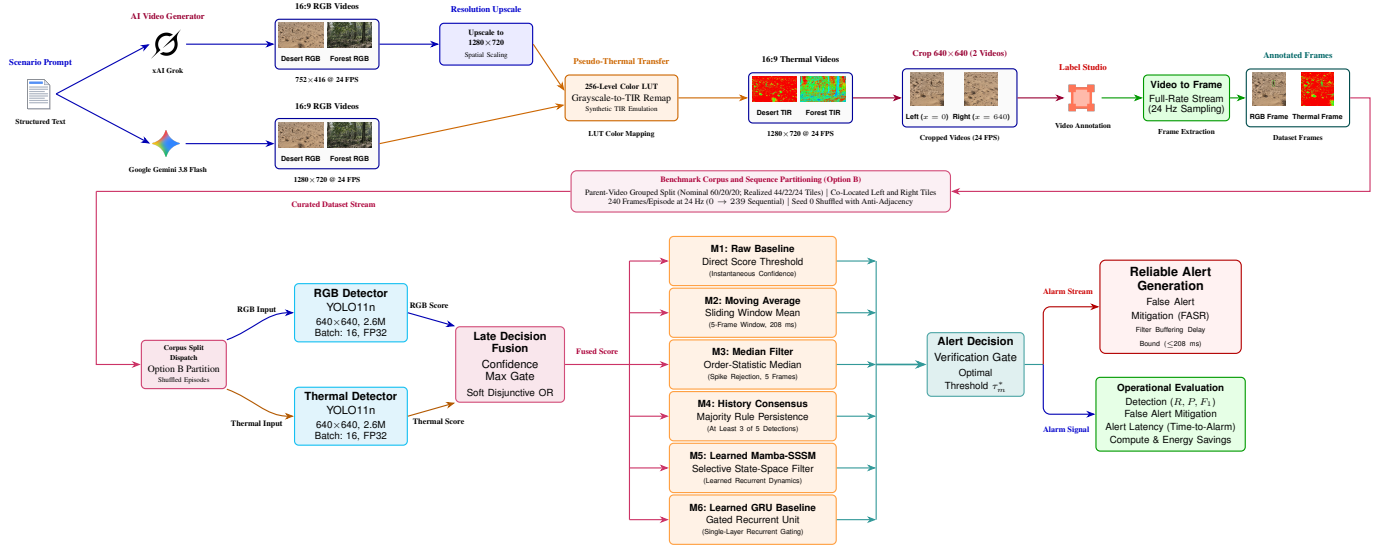


Fig. 1. System architecture of the MMSAR evaluation framework. Visible RGB video streams are synthesized from scenario text prompts using multimodal large language models (Google Gemini 3.8 Flash and xAI Grok with spatial upscaling to 1280×720). A 256-level lookup table (`melo_lut`) transfers pseudo-thermal styling to emulate synchronized thermal infrared streams. Dual streams undergo $2 \times$ square tiling (640×640), video-level bounding box annotation in Label Studio, and 24 Hz frame extraction. Curated sequences are partitioned into 240-frame episodes under Option B (parent-video grouped split) with non-adjacent shuffling to prevent spatial data leakage. At edge inference, parallel lightweight YOLO11n detectors process incoming RGB and thermal frames at 24 Hz ($T_s \approx 41.7$ ms), combined with a late fusion max gate ($s[n] = \max(c_{\text{RGB}}[n], c_{\text{Thermal}}[n])$) and evaluated across six lightweight temporal decision methods ($W = 5$ frames, 208 ms latency bound).

perimental setup covering desert, forest, and snow alpine biomes across synchronized RGB and pseudo-TIR video streams, evaluating fixed lightweight YOLO11n detectors with late fusion (Table I, Table II).

- 2) **Causal Decision Benchmark:** Empirical evaluation of six causal decision filters (raw thresholding, moving average, median filter, history consensus, Mamba-SSSM, and GRU) on alert-level metrics, multi-seed variance, and paired episode-level statistical tests (Table III, Fig. 2), establishing operational Pareto trade-offs between false alarm rejection, alert latency, and edge compute overhead.
- 3) **Window Sensitivity and Environmental Breakdown:** Empirical ablation of causal window length ($W \in \{1, 3, 5, 7\}$) on the validation split and per-biome false alert characterization across desert, forest, and snow terrain, demonstrating that finite impulse response (FIR) moving averages achieve the optimal operational trade-off between false alarm rejection and real-time responsiveness (Table IV, Fig. 3).

II. RELATED WORK

Multimodal aerial vision combines complementary visible and thermal bands to mitigate environmental extremes, as explored in ground datasets such as KAIST [4] and LLVIP [5]. For wilderness search and rescue (SAR), WiSARD [10] provides multi-viewpoint benchmarks, while Rudol and Doherty [1] demonstrated dual-band geolocation from unmanned aerial vehicles (UAVs). However, edge UAV deployment is tightly constrained by size, weight, power, and computing (SWaP-C) limits [2], where complex neural ensembles rapidly drain

battery reserves [11]. Single-stage detectors such as YOLO11n [8] deliver real-time frame rates, but per-frame detections fluctuate due to motion blur and terrain thermal crossover [6], generating false alarms that trigger wasteful hover maneuvers.

Classical sequential decision making addresses temporal fluctuations through statistical tests such as Wald’s sequential probability ratio test [12], radar sliding-window binary integration (M -of- N) [13], and running median or moving average filters [14]. More recently, selective state-space models such as Mamba [9] provide linear-time sequence modeling with recurrent state representations. While prior SAR literature emphasizes per-frame bounding box accuracy or spatial tracking, the problem of turning noisy multimodal scalar confidence streams into reliable mission-level alerts under strict edge latency budgets remains uncharacterized. This paper addresses this gap by benchmarking causal temporal decision rules on fixed multimodal detector streams.

III. METHODOLOGY

The MMSAR pipeline integrates dual-spectrum detection, late fusion, and causal temporal filtering (Fig. 1). Synchronized visible RGB frames $I_{\text{RGB}}[n]$ and thermal frames $I_{\text{Thermal}}[n]$ produce per-frame target confidences $c_{\text{RGB}}[n], c_{\text{Thermal}}[n] \in [0.0, 1.0]$. The upstream detectors and late fusion gate remain fixed across all experiments.

A. Video Ingestion and Real-Time Edge Budget

Autonomous UAV companion computers require frame-synchronous processing that matches sensor capture rates. Video streams are captured at $f_s = 24$ Hz, yielding a frame

period $T_s \approx 41.7$ ms. To avoid dropping frames or accumulating queue delay, processing must satisfy:

$$T_{\text{detect}} + T_{\text{fusion}} + T_{\text{filter}} \leq 41.7 \text{ ms}, \quad (1)$$

where $T_{\text{detect}} = \max(T_{\text{RGB}}, T_{\text{Thermal}})$ is parallel detection runtime, T_{fusion} is fusion latency, and T_{filter} is post-processing latency. We deploy two parallel YOLO11n models [8]: (1) **YOLO11n-RGB** and (2) **YOLO11n-Thermal**. Frames containing no target predictions yield confidence 0.0.

B. Late Fusion Max Gate

To prevent target dropouts when one sensor is occluded, detector confidences are combined using a late fusion max gate:

$$s[n] = \max(c_{\text{RGB}}[n], c_{\text{Thermal}}[n]) \in [0.0, 1.0]. \quad (2)$$

This max gate ensures that confident detections in either spectrum propagate forward without requiring pixel-level sensor registration.

C. Causal Temporal Post-Processing Methods

Direct thresholding of raw per-frame confidence produces high false-alarm rates due to brief clutter spikes. Downstream processing buffers the fused confidence stream across a causal sliding window of length $W = 5$ frames (208 ms at 24 Hz): $\mathbf{w}[n] = [s[n-4], \dots, s[n]]$. Each method uses an optimal decision threshold $\tau^* = \arg \max_{\tau} F_1(\tau)$ calibrated on the validation set $\mathcal{D}_{\text{val}}$ across candidate thresholds from 0.05 to 0.95 (step size 0.02) and held frozen during testing. *Protocol Lock*: No detector weights, temporal-model parameters, decision thresholds, or window lengths were selected or tuned using the test set.

We evaluate six causal decision strategies: **M1 (Baseline Raw)**: Instantaneous thresholding setting $y[n] = 1$ when $s[n] \geq \tau^*$ ($O(1)$ time, 0 B buffer). **M2 (Moving Average)**: Causal FIR mean $\bar{s}[n] = \frac{1}{W} \sum_{k=0}^{W-1} s[n-k]$, triggering an alert when $\bar{s}[n] \geq \tau^*$ ($O(1)$ time, 20 B buffer). **M3 (Median Filter)**: Order-statistic median $\tilde{s}[n] = \text{median}(\mathbf{w}[n])$, rejecting noise pulses shorter than 3 frames ($O(W)$ time, 20 B buffer). **M4 (History Consensus)**: Binary voting requiring at least 3 detections out of 5 frames ($O(W)$ time, 5 B buffer). **M5 (Learned Mamba-SSSM)**: Pure-PyTorch selective state-space model ($d_{\text{model}} = 16, d_{\text{state}} = 8$, 753 parameters, 512 B state buffer) updating a recurrent state from scalar confidences. **M6 (Learned GRU Baseline)**: Single-layer gated recurrent unit ($d_{\text{hidden}} = 16$, 1,681 parameters, 64 B state buffer) to benchmark whether conventional recurrent architectures capture temporal dependencies beyond linear low-pass filtering. Both learned models are trained on training partition confidence streams (44 tiles, 9,800 frames per modality) across five random seeds using binary cross-entropy, while decision thresholds τ^* are calibrated strictly on validation sequences.

TABLE I
CORPUS PARTITIONING AND EXPERIMENTAL SYSTEM SPECIFICATIONS

System Parameter	Design Specification / Value
Environments	Arid Desert (30 tiles), Temperate Forest (28 tiles), Snow-Covered Alpine (32 tiles)
Sensor Modalities	Synchronized visible RGB and pseudo-thermal infrared (TIR)
Raw Video Format	1280 × 720 at 24 FPS ($T_s \approx 41.7$ ms), 10.0 s nominal
Dual-Crop Tiling	2 × square tiles (640 × 640, left/right, $y = 40$)
Corpus Volume	90 tiles, 20,364 frames per modality (40,728 total frames; 42 positive tiles [5,620 frames], 48 negative distractor tiles [14,744 frames])
Episodic Partition	Option B parent-grouped (realized: 44 train [17 pos, 27 neg], 22 val [12 pos, 10 neg], 24 test [13 pos, 11 neg] tiles per modality)
Upstream Detectors	Dual YOLO11n (2.6M params, 6.5 GFLOPs at 640 × 640, FP32)
Training Setup	Batch 16, 60 epochs, SGD cosine schedule ($\text{lr}_0 = 0.01$)
Decision Window	Sliding causal window $W \in \{1, 3, 5, 7\}$ ($W = 5$ nominal; 208 ms)
Decision Strategies	Raw (M1), Moving Avg (M2), Median (M3), Consensus (M4), Mamba-SSSM (M5), GRU Baseline (M6)

D. Alert-Level Evaluation Metrics

To measure operational mission effectiveness, we report primary alert-level metrics:

- **False Alerts per Minute (FA/min)**: Rate of false alarm events triggered during non-target periods: $\text{FA/min} = K_{\text{FA}}/T_{\text{flight}}$, where an alarm event is defined as a contiguous sequence of positive detections ($y[n] = 1$).
- **Positive Target Recall (%)**: Proportion of ground-truth target tiles successfully detected by at least one confirmed alert event.
- **Alert Latency (Time-to-Alarm)**: Elapsed flight time (in frames and milliseconds) from target entrance in the camera field of view (FOV) to the first asserted alert ($y[n] = 1$). Crucially, this empirical metric captures both the algorithmic filter buffering delay ($\Delta t_{\text{buffer}} = (W - 1)T_s \approx 167$ ms, bounded by the 208 ms causal window at 24 Hz) and upstream detector confidence ramp-up as distant targets approach the UAV.
- **False Alarm Suppression Ratio (FASR)**: Percentage reduction in false alarm frames compared to the raw baseline:

$$\text{FASR} = 1 - \frac{\text{FP}_{\text{filter}}}{\text{FP}_{\text{raw}}}, \quad (3)$$

where $\text{FP}_{\text{filter}}$ and FP_{raw} are the total false-positive frames produced by the temporal filter and raw baseline, respectively.

IV. EXPERIMENTAL SETUP AND DATASET

Wilderness SAR dual-spectrum datasets remain scarce in the literature. As summarized in Table I, our controlled 3×2 experimental design balances modality and environment across three wilderness biomes:

- **Arid Desert**: Rocky terrain, sparse vegetation, and mid-day thermal crossover, where sun-heated rocks match human skin temperature.
- **Temperate Forest**: Dense canopies, dappled shadows, intermittent foliage occlusions, and wildlife distractors.
- **Snow-Covered Alpine**: High-albedo snow causing visible glare, contrasted against freezing backgrounds where winter clothing insulates body heat.

Raw aerial video is captured in 1280×720 resolution at 24 frames per second (FPS). To preserve ground sampling distance (GSD) without downsampling small human targets, a

TABLE II
FRAME-LEVEL UPSTREAM DETECTION PERFORMANCE AND MODALITY
PROGRESSION ON HELD-OUT TEST SPLIT (FIXED DETECTORS,
THRESHOLD $\tau = 0.50$)

Sensing Configuration / Environment	Precision	Recall	F1-Score
<i>Arid Desert Biome</i>			
YOLO11n-RGB	0.970	0.868	0.916
YOLO11n-Thermal	0.997	0.851	0.918
Late Fusion Max Gate	0.970	0.942	0.956
<i>Temperate Forest Biome</i>			
YOLO11n-RGB	0.922	0.444	0.599
YOLO11n-Thermal	0.915	0.318	0.471
Late Fusion Max Gate	0.907	0.521	0.662
<i>Snow/Alpine Biome</i>			
YOLO11n-RGB	0.997	0.523	0.686
YOLO11n-Thermal	0.952	0.206	0.338
Late Fusion Max Gate	0.978	0.537	0.693
<i>Modality Progression (Overall Test Set, 5,570 frames)</i>			
YOLO11n-RGB (Raw, $\tau = 0.50$)	0.967	0.637	0.768
YOLO11n-Thermal (Raw, $\tau = 0.50$)	0.975	0.492	0.654
Late Fusion Max Gate (Raw, $\tau = 0.50$)	0.958	0.693	0.804
Late Fusion Max + Moving Avg ($W = 5, \tau^* = 0.57$)	0.978	0.650	0.781

Bold indicates best result per group. Max-pooling maximizes recall while temporal smoothing maximizes precision by rejecting noise spikes.

$2 \times$ dual-crop extraction produces disjoint 640×640 square tiles (left tile $x \in [0, 640]$ and right tile $x \in [640, 1280]$, $y = 40$). Because target persons appear within a single spatial tile in positive parent captures while the companion tile captures ambient terrain, classifying tiles by bounding box presence yields 30 desert tiles (13 positive, 17 negative), 28 forest tiles (14 positive, 14 negative), and 32 snow alpine tiles (15 positive, 17 negative). Across the 90 tiles per modality (42 positive, 48 negative distractor tiles; 180 synchronized streams), the corpus comprises 20,364 frames at 24 Hz per modality (40,728 frames total: 5,620 target frames and 14,744 negative distractor frames per modality). Sequences are annotated in Label Studio using keyframe interpolation, yielding precise ground-truth bounding boxes.

Dataset partitioning is conducted strictly at the parent-video level under Option B (Parent-Video Grouped Episodic Partitioning) to prevent spatial or temporal data leakage between derived sibling tiles. Sibling left and right tiles are assigned to the same partition. The realized partition allocates 44 training tiles (17 positive, 27 negative; 9,800 frames per modality), 22 validation tiles (12 positive, 10 negative; 4,994 frames per modality), and 24 testing tiles (13 positive, 11 negative; 5,570 frames per modality). In the held-out test split, each biome contributes exactly 8 tiles (desert: 4 positive, 4 negative; forest: 4 positive, 4 negative; snow alpine: 5 positive, 3 negative). Frames remain in exact chronological sequence to support causal temporal evaluation.

V. RESULTS AND DISCUSSION

A. Upstream Detection Across Modalities and Biomes

We first evaluate the upstream frame-level detection performance of the two fixed YOLO11n models and the late fusion max gate on the held-out test split under standard thresholding ($\tau = 0.50$). Table II summarizes precision, recall, and F1-score across all three biomes.

The upstream modalities exhibit distinct physical failure modes across environments: visible RGB provides sharp local-

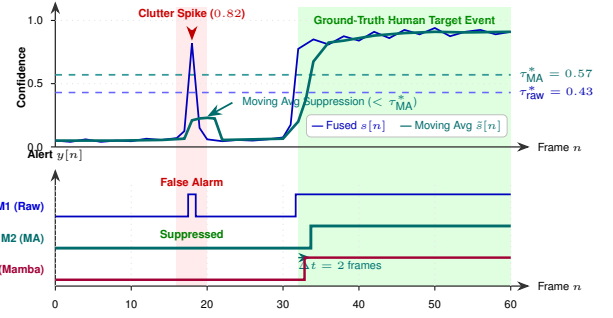


Fig. 2. Temporal sequence trace from an actual held-out test clip: (top) upstream fused confidence $s[n]$ exhibiting an isolated clutter spike at frame 18 followed by a sustained ground-truth target event from frame 32; (bottom) causal binary alert decisions comparing baseline thresholding M1 (spurious alert triggered at frame 18), five-frame moving average M2 (clutter spike completely suppressed, target confirmed with 2-frame latency), and Mamba-SSSM M5.

ization under open sunlight but suffers severe recall dropouts under dense forest canopy shadows (recall drops to 44.4%); conversely, the thermal stream is attenuated by thick foliage canopy (forest recall drops to 31.8%, though maintaining 91.5% precision on visible target patches) and incurs false positives on sun-heated desert rocks. Combining both streams via the late fusion max gate ($s[n] = \max(c_{\text{RGB}}[n], c_{\text{Thermal}}[n])$) mitigates these unimodal dropouts, elevating forest frame-level recall to 52.1% (a +7.7% improvement over RGB and +20.3% over thermal alone) and achieving 94.2% frame-level recall in desert. However, because max-pooling forwards the highest confidence score, it also propagates isolated noise spikes from both sensors into downstream decision buffers, producing false alarms on negative distractor scenes. Multi-seed detector evaluation across three random seeds (seeds 0, 42, 1234) confirmed stable upstream convergence ($\text{mAP}_{50} = 0.985 \pm 0.008$, $\text{mAP}_{50-95} = 0.518 \pm 0.017$). This motivates lightweight causal temporal filtering.

B. Comparative Evaluation of Causal Decision Methods

Across the held-out test split, causal temporal post-processing substantially suppresses false alarm events, revealing distinct operational trade-offs across decision methods. Table III reports the primary benchmark results across the six decision strategies, evaluated under validation-calibrated thresholds τ_m^* with 95% confidence intervals across evaluation episodes.

The five-frame moving average (M2) achieves the strongest clutter suppression among non-learned methods, reducing false alerts by 77% (from 9.0 down to 2.1 FA/min) and achieving a 65.3% frame-level false alarm suppression ratio (FASR, reducing false-positive frames from 75 down to 26), with an alert latency of 592.3 ms and 84.6% target recall (11 of 13 positive test tiles detected, missing one canopy-occluded forest target and one distant snow target wearing insulating winter clothing). A paired Wilcoxon signed-rank test across the 24 held-out test episodes confirms that M2's reduction in false positive frames ($W = 2.0, p = 0.0056$) and false alert

TABLE III
COMPARATIVE BENCHMARK OF CAUSAL TEMPORAL DECISION METHODS ON HELD-OUT TEST SEQUENCES ($W = 5$ FRAMES / 208 MS AT 24 HZ)

Decision Method	Val τ^*	Alert Recall (%)	Frame F_1 -Score [95% CI]	False Alerts / min	Alert Latency (Time-to-Alarm)	Filter Runtime / Frame
M1: Baseline Raw Thresholding	0.43	92.3% (12/13)	0.813 [0.776, 0.847]	9.0	188.9 ms (4.5 frames)	0.02 μs ($O(1)$, 0 B state)
M2: Five-Frame Moving Average	0.57	84.6% (11/13)	0.781 [0.738, 0.822]	2.1 [†]	592.3 ms (14.2 frames)	0.30 μ s ($O(1)$, 20 B state)
M3: Five-Frame Median Filter	0.49	92.3% (12/13)	0.805 [0.767, 0.841]	3.6	291.7 ms (7.0 frames)	0.42 μ s ($O(W)$, 20 B state)
M4: Five-Frame History Consensus	0.49	92.3% (12/13)	0.802 [0.764, 0.838]	3.6	313.9 ms (7.5 frames)	0.63 μ s ($O(W)$, 5 B state)
M5: Learned Mamba-SSSM (5 seeds)	0.73	87.7% $\pm$ 4.2%	0.717 $\pm$ 0.040 [0.672, 0.758]	1.7 $\pm$ 0.6	587.5 ms (14.1 frames)	150.59 μ s ($O(d_{\text{model}}d_{\text{state}})$, 512 B)
M6: Learned GRU Baseline (5 seeds)	0.50	86.2% $\pm$ 3.4%	0.822 $\pm$ 0.007 [0.785, 0.856]	3.7 $\pm$ 0.3	577.3 ms (13.9 frames)	21.80 μ s ($O(d_{\text{hidden}}^2)$, 64 B)

Bold indicates best metric in column. [†]M2 achieves the lowest FA/min among non-learned methods; difference vs. M5 is not statistically significant ($p = 0.4922$).

events ($W = 2.0, p = 0.0087$) over baseline M1 is statistically significant.

When mission priorities favor rapid alert triggering and maximum target sensitivity over absolute clutter rejection, order-statistic median filtering (M3) and history consensus (M4) establish an attractive alternative: both preserve 92.3% target recall (12/13 tiles detected) with faster alert latency (291.7 ms and 313.9 ms), while still achieving a 60% false alert reduction (3.6 FA/min, 32.0% FASR). Mathematically, for any odd window length $W = 2k + 1$ (here $W = 5$), the order-statistic condition $\text{median}(s[n-4], \dots, s[n]) \geq \tau$ is logically equivalent to requiring at least $k+1 = 3$ samples $\geq \tau$, matching the consensus rule of M4 on all steady-state frames ($n \geq 4$). They differ only during the initial 2-frame buffer fill-up, producing identical false alarm counts (14 events, 51 FP frames) with a minor 22.2 ms latency difference. Wilcoxon testing confirms M2 achieves significantly fewer false alarm frames than M3 ($W = 0.0, p = 0.0047$) and fewer false alert events ($W = 0.0, p = 0.0143$), delineating the clear trade-off between M2’s stronger smoothing and M3’s early detection sensitivity.

In comparison, learned sequence models do not establish a compelling operational advantage over simple low-pass filtering on this 1D scalar stream. While M5 Mamba-SSSM achieves numerically lower false alerts (1.7 $\pm$ 0.6 FA/min), an episode-level paired Wilcoxon test confirms that the difference between M2 and M5 is not statistically significant ($W = 5.0, p = 0.4922$). Meanwhile, M5 requires 500 $\times$ higher execution latency (150.59 μ s vs. 0.30 μ s), 25 $\times$ more state memory (512 B vs. 20 B), and yields significantly lower episode F_1 ($W = 8.0, p = 0.0150$). Similarly, while M6 GRU achieves the highest frame-level F_1 (0.822 $\pm$ 0.007), the difference in episode F_1 relative to M2 is not statistically significant ($W = 15.0, p = 0.0597$), while M6 incurs significantly more false alert events (3.7 $\pm$ 0.3 FA/min, $W = 0.0, p = 0.0178$) and 73 $\times$ greater runtime (21.80 μ s). Because transient clutter spikes are isolated impulses, parameterized recurrent models retain lingering hidden activations, whereas linear FIR averaging provides immediate $1/W = 0.20$ attenuation of isolated noise pulses without state delays or training overhead.

C. Environmental Robustness Across Biomes

To examine whether temporal filtering performs consistently across diverse search environments, Table IV stratifies performance across the three held-out test biomes (8 tiles per biome, 5,570 frames total). Across all terrains, five-frame moving

TABLE IV
ENVIRONMENT-STRATIFIED CAUSAL DECISION PERFORMANCE ON HELD-OUT TEST SPLIT (8 TILES PER BIOME, 5,570 TOTAL FRAMES)

Environment	Decision Method	Val τ^*	Alert Recall (%)	FA / min	FP Frames
Arid Desert	M1: Raw Baseline	0.43	100.0%	10.50	32
	M2: Moving Average	0.57	100.0%	1.50	10
	M3: Median Filter	0.49	100.0%	3.00	20
	M4: History Consensus	0.49	100.0%	3.00	20
Temperate Forest	M1: Raw Baseline	0.43	100.0%	12.49	34
	M2: Moving Average	0.57	75.0%	3.33	13
	M3: Median Filter	0.49	100.0%	4.99	21
	M4: History Consensus	0.49	100.0%	4.99	21
Snow/Alpine	M1: Raw Baseline	0.43	80.0%	4.50	9
	M2: Moving Average	0.57	80.0%	1.50	3
	M3: Median Filter	0.49	80.0%	3.00	10
	M4: History Consensus	0.49	80.0%	3.00	10

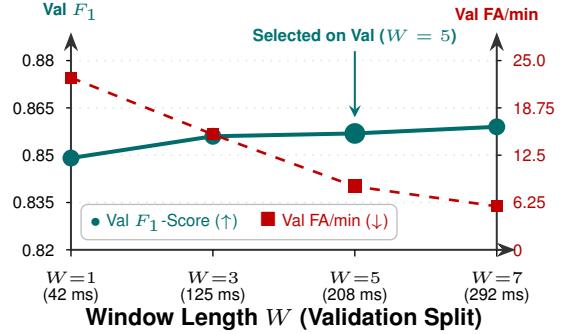


Fig. 3. Temporal window length ablation ($W \in \{1, 3, 5, 7\}$) for the moving average filter on the validation split, showing F_1 -score ($\uparrow$) and False Alerts/min ($\downarrow$) across causal latency bounds (42 ms to 292 ms at 24 Hz). $W = 5$ achieves the optimal trade-off on validation sequences, cutting false alerts by 63% with minimal latency.

average (M2) substantially lowers false alert rates compared to instantaneous baseline thresholding (M1): in arid desert, false alerts drop from 10.50 to 1.50 FA/min (an 85.7% reduction); in temperate forest, false alerts drop from 12.49 to 3.33 FA/min (a 73.3% reduction); and in snow alpine terrain, false alerts drop from 4.50 to 1.50 FA/min (a 66.7% reduction). While canopy occlusions in dense forest slightly reduce frame-level target overlap under heavy smoothing (alert recall 75.0% [3/4 tiles] for M2 vs. 100.0% [4/4 tiles] for order-statistic median M3, and 80.0% [4/5 tiles] in snow), M2 achieves the lowest false alert rate across all three biomes, confirming robust environmental transferability.

D. Temporal Window Length Sensitivity

A window of $W = 5$ frames (208 ms) achieves the optimal operating trade-off on validation sequences, cutting false alerts from 22.8 down to 8.4 FA/min (a 63.3% reduction) while maintaining high $F_1 = 0.857$ within a 208 ms

latency bound (Fig. 3). Ablating $W \in \{1, 3, 5, 7\}$ frames (41.7 ms to 291.7 ms at 24 Hz) on the validation partition demonstrates that $W = 1$ suffers from high clutter sensitivity (22.8 FA/min, $F_1 = 0.849$). Increasing to $W = 3$ smooths high-frequency noise, decreasing false alerts to 15.3 FA/min ($F_1 = 0.856$). At $W = 5$, the filter suppresses clutter by over 63% with responsive target acquisition. Expanding further to $W = 7$ yields diminishing returns (5.8 FA/min) while increasing buffer latency to 292 ms. Thus, $W = 5$ was selected on validation and held frozen for testing.

E. Limitations

Several constraints bound our findings. First, pseudo-thermal infrared streams are generated via a global grayscale lookup table rather than radiometric sensor physics; high-albedo snow renders as the warmest region in frame, inverting thermal contrast relative to physical snowpack. Second, the benchmark evaluates 24 held-out test tiles (5,570 frames per modality under Option B parent-video grouping, 8 tiles per biome with 13 positive and 11 negative tiles). On alert recall, M2 detected 11 of 13 positive test tiles (84.6%), missing one distant target under dense forest canopy and one in snow where insulated clothing reduced thermal emissions. Third, hyperparameter $W = 5$ was selected on validation sequences to balance delay and clutter attenuation; mission-specific flight speeds may require adaptive window scaling. Fourth, clips capture nominal aerial trajectories (30–50 m AGL at 8–15 m/s) in clear conditions; violent turbulence and degraded weather (heavy fog, blizzard snowfall) require further study. Finally, timing benchmarks reflect desktop GPU execution; embedded Jetson Orin Nano flight validation is planned for subsequent work.

VI. CONCLUSION

This paper investigated lightweight multimodal person detection with causal temporal post-processing for UAV wilderness SAR across desert, forest, and snow biomes. Holding upstream YOLO11n detectors and the late fusion max gate fixed, we benchmarked six causal decision methods under validation-calibrated thresholds. Causal temporal post-processing establishes a clear operational Pareto trade-off: a five-frame moving average achieves the lowest false alert rate among non-learned methods (2.1 FA/min, a 77% reduction vs. baseline, 65.3% FASR) with 84.6% target recall at sub-microsecond latency (0.30 μ s runtime, 20-byte buffer), while five-frame median filtering preserves 92.3% recall with responsive 291.7 ms alert latency at 3.6 FA/min. Parameterized sequence models offer no operational advantage over linear moving averages: Mamba-SSSM yields statistically indistinguishable false alerts ($p = 0.4922$) while GRU incurs significantly more false alert events ($p = 0.0178$), both imposing $73\times$ to $500\times$ higher computational latency and state memory overhead. Lightweight causal post-processing provides a computationally negligible, training-free layer for preventing wasteful UAV loiters and conserving telemetry bandwidth in autonomous aerial search operations.

AI USE DISCLAIMER

Generative artificial intelligence (AI) tools were used in this research for synthetic training video generation (Google Gemini 3.8 Flash and xAI Grok) as well as AI-assisted manuscript editing and proofreading. All experimental designs, model training, algorithmic code, post-processing pipelines, quantitative evaluations, and scientific interpretations were developed and verified by the authors.

DATA AND CODE AVAILABILITY

Simulation frameworks, snippet harvesting tools, dataset split manifests, and PyTorch temporal decision implementations are available at: <https://github.com/IamOumarIbrahim/multi-modal-detection>.

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REFERENCES

- [1] P. Rudol and P. Doherty, "Human body detection and geolocation for UAV search and rescue operations using color and thermal imagery," in *Proceedings of the IEEE Aerospace Conference*, 2008, pp. 1–8.
- [2] R. Silvagni, A. Tonoli, E. Zenerino, and M. Chiaberge, "Multipurpose UAV for search and rescue operations in mountain avalanche events: Design considerations," *Geomatics, Natural Hazards and Risk*, vol. 8, no. 1, pp. 18–33, 2016.
- [3] C. Kyrkou and T. Theodoridis, "EmergencyNet: Efficient aerial image classification for drone-based emergency monitoring using atrous convolutional feature fusion," *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, vol. 13, pp. 1687–1699, 2020.
- [4] S. Hwang, J. Park, N. Kim, Y. Choi, and I. S. Kweon, "Multispectral pedestrian detection: Benchmark dataset and baseline," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2015, pp. 1037–1045.
- [5] X. Jia, C. Zhu, M. Li, W. Tang, and W. Zhou, "LLVIP: A visible-infrared paired dataset for low-light vision," in *Proceedings of the IEEE/CVF International Conference on Computer Vision Workshops (ICCVW)*, 2021, pp. 3489–3497.
- [6] D. Portmann, S. Lynen, M. Chli, and R. Siegwart, "People detection and tracking from aerial thermal views," in *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA)*, 2014, pp. 1794–1800.
- [7] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You only look once: Unified, real-time object detection," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016, pp. 779–788.
- [8] G. Jocher and J. Qiu, "Ultralytics YOLO11," 2024. [Online]. Available: <https://github.com/ultralytics/ultralytics>
- [9] A. Gu and T. Dao, "Mamba: Linear-time sequence modeling with selective state spaces," in *Proceedings of the First Conference on Language Modeling (COLM)*, 2024, also arXiv:2312.00752.
- [10] D. Broyles, C. R. Hayner, and K. Leung, "WiSARD: A labeled visual and thermal image dataset for wilderness search and rescue," in *Proceedings of the IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2022, pp. 9467–9474.
- [11] V. Lygouras, N. Santavas, A. Taitzoglou, K. Tarchanidis, A. Mitropoulos, and A. Gasteratos, "Unsupervised human detection with an embedded vision system on a fully autonomous UAV for search and rescue operations," *Sensors*, vol. 19, no. 16, p. 3542, 2019.
- [12] A. Wald, "Sequential tests of statistical hypotheses," *The Annals of Mathematical Statistics*, vol. 16, no. 2, pp. 117–186, 1945.
- [13] M. A. Richards, *Fundamentals of Radar Signal Processing*, 2nd ed. New York, NY, USA: McGraw-Hill Education, 2014.
- [14] L. R. Rabiner, M. R. Sambur, and C. E. Schmidt, "Applications of a nonlinear smoothing algorithm to speech processing," *IEEE Transactions on Acoustics, Speech, and Signal Processing*, vol. 23, no. 6, pp. 552–557, 1975.